

Frontotemporal Dementia And/Or Amyotrophic Lateral Sclerosis 2 (FTDALS2): Comprehensive Disease Characteristics Report

Disease: Frontotemporal Dementia And/Or Amyotrophic Lateral Sclerosis 2 (FTDALS2)
MONDO ID: MONDO:0014395 · **OMIM:** #615911 · **Causal gene:** *CHCHD10* (HGNC:15559)
Category: Mendelian (autosomal dominant)

Summary

Frontotemporal Dementia and/or Amyotrophic Lateral Sclerosis 2 (FTDALS2; OMIM #615911, MONDO:0014395) is a rare, adult/late-onset, autosomal dominant neurodegenerative disorder caused by heterozygous gain-of-function missense mutations in *CHCHD10*, a gene encoding a small mitochondrial intermembrane-space protein. It was defined by the discovery of the founding p.Ser59Leu (c.176C>T) mutation in a large French family exhibiting a strikingly multisystem "mitochondrial" phenotype that combined motor neuron disease, frontotemporal-dementia-like cognitive decline, cerebellar ataxia, and mitochondrial myopathy with ragged-red/COX-negative fibres and multiple mitochondrial DNA (mtDNA) deletions [PMID: 24934289]. FTDALS2 sits within a broader **CHCHD10 spectrum** that spans classical ALS, FTD-ALS, isolated mitochondrial myopathy/cardiomyopathy, late-onset spinal motor neuronopathy (SMAJ, Spinal Muscular Atrophy Jokela type), and Charcot-Marie-Tooth disease type 2, with a strong allele-specific genotype-phenotype correlation [PMID: 37021679; 25428574].

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Mechanistically, mutant CHCHD10 misfolds and aggregates, disassembles the MICOS (mitochondrial contact site and cristae organizing system) complex, collapses cristae junctions, and destabilizes the mitochondrial genome — explaining the accumulation of deleted mtDNA in patient muscle [PMID: 26666268]. Downstream, mutant protein activates the OMA1 peptidase, which cleaves the long form of the fusion protein OPA1 (L-OPA1), triggers the

mitochondrial integrated stress response (mtISR), impairs mitochondrial axonal transport, and drives cytoplasmic accumulation of TDP-43, the pathological hallmark shared with sporadic ALS/FTLD-TDP [PMID: 32338760;

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The mechanism is best described as a **toxic gain-of-function / dominant-negative**, not simple haploinsufficiency; the degree of MICOS disruption tracks clinical severity [PMID: 30092269].

CHCHD10 mutations are an **ultra-rare** cause of ALS/FTD (~0.4% of cohorts), and careful population-genetics work is essential to separate genuinely pathogenic ultra-rare alleles (p.Ser59Leu, p.Arg15Leu) from historically-reported benign polymorphisms such as p.Pro34Ser (gnomAD allele frequency ~0.4%, far too common to be causal) [PMID: 27056076;

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There is **no disease-specific therapy**; management follows ALS/FTD symptomatic standards (riluzole, edaravone, multidisciplinary supportive care) [PMID: 42113599;

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Experimental precision strategies targeting the MICOS/mitochondrial-transport axis (e.g., nifuroxazide), the integrated stress response, and metabolic/creatine buffering are under active investigation in yeast, knock-in mouse, and patient iPSC-derived motor neuron models [PMID: 39478664;

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Key Findings

Finding 1 — *CHCHD10* is the causal gene (autosomal dominant)

FTDALS2 is caused by heterozygous mutations in **CHCHD10** (coiled-coil-helix-coiled-coil-helix domain containing 10, chromosome 22q11.23, HGNC:15559), encoding a mitochondrial intermembrane-space protein. The founding p.Ser59Leu mutation was identified in a large French family with mitochondrial myopathy associated with motor neuron disease: *"We reported patients, carrying the p.Ser59Leu heterozygous mutation in CHCHD10, from a large*

family with a mitochondrial myopathy associated with motor neuron disease (MND)" [PMID: 30874923]. The allelic spectrum was subsequently extended across ALS, FTD-ALS, and milder syndromes; for example, a distinct milder allele defines a lower motor neuron syndrome: "Mutation c.197G>T p.G66V in CHCHD10 is the cause of the lower motor neuron syndrome LOSMoN/SMAJ" [PMID: 25428574]. Inheritance is autosomal dominant.

Finding 2 — Core lesion: MICOS disassembly, cristae collapse, mtDNA instability

CHCHD10 resides within the MICOS complex together with mitofilin (MIC60), CHCHD3, and CHCHD6. Mutant CHCHD10 disassembles MICOS and collapses cristae junctions: *"CHCHD10 resides with mitofilin, CHCHD3 and CHCHD6 within the 'mitochondrial contact site and cristae organizing system' (MICOS) complex. CHCHD10 mutations lead to MICOS complex disassembly and loss of mitochondrial cristae with a decrease in nucleoid number and nucleoid disorganization" [PMID: 26666268]. This links directly to mtDNA instability: "Repair of the mitochondrial genome after oxidative stress is impaired in CHCHD10 mutant fibroblasts and this likely explains the accumulation of deleted mtDNA molecules in patient muscle" [PMID: 26666268]. Importantly, the **degree of MICOS disruption correlates with disease severity**: "Loss of MICOS complex integrity and mitochondrial damage, but not TDP-43 mitochondrial localisation, are likely associated with severity of CHCHD10-related diseases" [PMID: 30092269].*

Finding 3 — Toxic gain-of-function → OMA1/OPA1 cleavage → mtISR → cytoplasmic TDP-43

The pathogenic mechanism is a tissue-specific toxic gain-of-function with dominant-negative activity, not haploinsufficiency. A knock-in mouse bearing the mouse-equivalent S59L (S55L) mutation was generated *"to investigate the pathogenic mechanisms of CHCHD10 ... harboring the mouse-equivalent of a disease-associated human S59L mutation, S55L in the endogenous mouse gene,"* demonstrating a tissue-specific toxic gain-of-function and mitochondrial stress response [PMID: 30877432]. The downstream cristae-shaping mechanism is OMA1-mediated: *"C2/C10 DKO mice have disrupted mitochondrial cristae, because of cleavage of the mitochondrial-shaping protein long form of OPA1 (L-OPA1) by the stress-induced peptidase OMA1,"* and these mice *"partially phenocopied mutant C10 KI mice with the development of cardiomyopathy and activation of the integrated mitochondrial integrated stress response in affected tissues" [PMID: 32338760]. The link to TDP-43 pathology is direct: "FTD/ALS-associated mutations (R15L and S59L) exhibit loss of function phenotypes in C. elegans genetic complementation assays and dominant negative activities in mammalian systems, resulting in*

mitochondrial/synaptic damage and cytoplasmic TDP-43 accumulation" [PMID: 28585542]. Insoluble CHCHD10 co-aggregates with phospho-TDP-43 and correlates with insoluble TDP-43 in FTLT-TDP brains [PMID: 35787294].

Finding 4 — Multisystem late-onset clinical phenotype

In the founding p.Ser59Leu (c.176C>T) family, *"We report a large family with a late-onset phenotype including motor neuron disease, cognitive decline resembling frontotemporal dementia, cerebellar ataxia and myopathy"* [PMID: 24934289]. Muscle biopsy findings were characteristically mitochondrial: *"In all patients, muscle biopsy showed ragged-red and cytochrome c oxidase-negative fibres with combined respiratory chain deficiency and abnormal assembly of complex V,"* and *"The multiple mitochondrial DNA deletions found in skeletal muscle revealed a mitochondrial DNA instability disorder"* [PMID: 24934289]. Patient fibroblasts showed respiratory chain deficiency, mitochondrial ultrastructural alterations, and fragmentation of the mitochondrial network; overexpression of mutant CHCHD10 in HeLa cells caused loss, disorganization, and dilatation of cristae.

Finding 5 — Rarity, severity gradient, and pathology staging

CHCHD10 mutations are a rare cause of ALS/FTD-ALS. Screening of 499 Chinese ALS patients found *"The mutation frequency of CHCHD10 (0.4 %, 2/487) in a Chinese SALS population"* [PMID: 27056076], and some variants have a *"controversial role in ALS"* [PMID: 28318595]. Variant-dependent severity is well established: p.Ser59Leu causes severe FTD-ALS with mtDNA instability, whereas the SMAJ phenotype is mild — *"patients presenting with SMAJ phenotype have neither mitochondrial myopathy nor mtDNA instability"* [PMID: 30092269];

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In knock-in mice, *"Mitochondrial defect in muscle precedes neuromuscular junction degeneration and motor neuron death in CHCHD10"* [PMID: 30874923], establishing a muscle → NMJ → motor neuron staging.

Finding 6 — Population genetics separates pathogenic from benign variants

A direct gnomAD v4 query (CHCHD10, ENSG00000250479, chr22:23,765,834–23,767,972, GRCh38) confirmed that pathogenic FTD-ALS alleles are ultra-rare: **p.Ser59Leu** allele frequency (AF) = 6.9×10^{-7} (1 of 1,446,818 exome alleles); **p.Arg15Leu** AF = 0 (0 of 949,590); **p.Ala35Asp** AF = 6.1×10^{-6} ; the Finnish SMAJ founder **p.Gly66Val** AF = 2.1×10^{-6} (3 alleles). By stark contrast, **p.Pro34Ser**, reported in early ALS/FTD studies, has AF = 0.0043 (5,542 exome

alleles; ~0.23% in genomes) — far above any plausible disease-allele frequency, indicating it is a benign/likely-benign common polymorphism (ACMG BA1/BS1). This population evidence explains the literature's caution that CHCHD10 has *"a controversial role in ALS"* [PMID: 28318595] — the controversy largely reflects benign common variants being conflated with true ultra-rare pathogenic alleles.

Finding 7 — Strong allele-specific genotype–phenotype correlation

Distinct CHCHD10 alleles map to distinct clinical syndromes: *"dominant mutations in the mitochondrial protein CHCHD10 (p.R15L and p.S59L) and its paralog CHCHD2 (p.T61I) were shown to cause familial amyotrophic lateral sclerosis (ALS) and Parkinson's disease (PD), respectively"* [PMID: 37021679]. Further: *"Different mutations in CHCHD10 cause additional neuromuscular disorders, including the lower motor neuron disease Spinal Muscular Atrophy Jokela type (SMAJ) (p.G66V) and autosomal dominant isolated mitochondrial myopathy (IMMD) (p.G58R)"* [PMID: 37021679]. The unifying mechanism is toxic misfolding: *"mitochondrial dysfunction may drive ALS and PD pathogenesis by a gain of function mechanism, driven by protein misfolding of CHCHD2 and CHCHD10 into toxic species"* [PMID: 37021679]. CHCHD10 is an intrinsically disordered/low-complexity protein that heterodimerizes with its paralog CHCHD2 [PMID: 35791387;

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Finding 8 — No disease-specific therapy; emerging precision strategies

No approved CHCHD10-specific therapy exists; management follows ALS/FTD symptomatic standards (riluzole — modest survival benefit; edaravone — narrow eligibility; multidisciplinary care) [PMID: 42113599;

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Experimental precision approaches are emerging. A yeast-based repurposing screen identified nifuroxazide: *"nifuroxazide rescues mitochondrial network fragmentation and cristae abnormalities in CHCHD10^{S59L}/+ patient fibroblasts. This molecule also decreases caspase-dependent death of human CHCHD10^{S59L}/+ induced pluripotent stem cell-derived motor neurons,"* and *"Its benefits involve KIF5B-mediated mitochondrial transport enhancement, evidenced by increased axonal movement and syntaphilin degradation in patient-derived motor neurons"* [PMID: 39478664]. Metabolic dysregulation is a downstream feature and potential biomarker/target: *"CHCHD10 p.G66V dysregulates energy metabolism, leading to altered redox*

balance and energy buffering by creatine metabolism," and "we report the first homozygous *CHCHD10* patient, and show that the variant dosage dictates the severity of the motor neuron disease in SMAJ" [PMID: 40400037].

Report by Section

1. Disease Information

FTDALS2 is a Mendelian, autosomal dominant, adult/late-onset neurodegenerative disorder within the ALS–FTD spectrum, caused by *CHCHD10* mutations and distinguished by prominent mitochondrial features (myopathy with ragged-red/COX-negative fibres, mtDNA instability) alongside motor neuron disease, FTD-like cognitive decline, and cerebellar ataxia [PMID: 24934289].

- **Key identifiers:** OMIM #615911; MONDO:0014395; gene *CHCHD10* (HGNC:15559; NCBI Gene 400916; Ensembl ENSG00000250479); chromosome 22q11.23. ICD-10 mapping is via G31.0 (frontotemporal dementia) and G12.21 (ALS); ICD-11 in the 8B60/8B00 range. MeSH: Frontotemporal Dementia; Amyotrophic Lateral Sclerosis.
- **Synonyms / alternative names:** FTDALS2; "FTD-ALS type 2"; part of the "CHCHD10 spectrum" / "CHCHD10-related disease"; historically described as "mitochondrial myopathy with motor neuron disease." Related spectrum entities include SMAJ (Jokela-type spinal muscular atrophy), isolated mitochondrial myopathy (IMMD), and CMT2.
- **Data source type:** Information is derived from aggregated disease-level resources (OMIM, Orphanet, ClinVar, gnomAD) and primary literature (family pedigrees, cohort screens, model organisms), not from EHR-level individual patient records.

2. Etiology

- **Causal factors:** Genetic. Heterozygous gain-of-function missense mutations in *CHCHD10* [PMID: 24934289;

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No infectious or purely environmental cause.

- **Genetic risk factors:** The causal variants are ultra-rare dominant missense alleles — principally **p.Ser59Leu (c.176C>T)** and **p.Arg15Leu** for the ALS/FTD-ALS phenotype [PMID: 37021679]. Allele-specific: p.G66V → SMAJ; p.G58R → isolated mitochondrial myopathy. The

paralog *CHCHD2* (p.T61I) causes Parkinson's disease, and CHCHD2/CHCHD10 heterodimerize [PMID: 37021679;

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- **Genetic pseudo-risk / benign confounders:** p.Pro34Ser is a common benign polymorphism (gnomAD AF ~0.4%) historically misattributed as a risk allele (ACMG BA1/BS1) — a key caution for variant interpretation (Finding 6) [PMID: 28318595].
- **Environmental / lifestyle risk factors:** Age (late-onset) is the dominant non-genetic factor; no established toxin, occupational, or lifestyle risk factor is specific to FTDALS2.
- **Protective factors:** No validated genetic or environmental protective factors are established for FTDALS2. (Not available.)
- **Gene–environment interactions:** Not characterized for FTDALS2 specifically. Oxidative stress exacerbates the mtDNA-repair defect in mutant cells, suggesting an inferred (not demonstrated) sensitization to oxidative-stress environments [PMID: 26666268].

3. Phenotypes

Phenotype	Type	Onset / severity / course	Frequency	Suggested HPO
Motor neuron disease (upper + lower) / ALS	Clinical sign	Adult/late-onset; progressive	Core, in founding family	HP:0007354 (ALS); HP:0002355; HP:0007289
Frontotemporal-dementia-like cognitive decline	Behavioral/ cognitive	Late-onset; progressive	Core	HP:0002145 (Frontotemporal dementia); HP:0100543
Cerebellar ataxia	Clinical sign	Late-onset; progressive	Present in founding family	HP:0001251 (Ataxia); HP:0002070
Mitochondrial myopathy (ragged-red, COX-negative fibres)	Lab/ histopathology	Adult; progressive	Core in FTDALS2	HP:0003198 (Myopathy); HP:0003200; HP:0008314
Respiratory chain deficiency / abnormal complex V assembly	Laboratory abnormality	—	Core	HP:0011922; HP:0003287
Multiple mtDNA deletions in muscle	Laboratory abnormality	—	Core (S59L)	HP:0003689 (Multiple mitochondrial DNA deletions)
Cardiomyopathy	Clinical sign	Variable	Spectrum-dependent	HP:0001638
Peripheral neuropathy (CMT2-like)	Clinical sign	Spectrum-dependent	Allele-dependent	HP:0009830

Evidence: "late-onset phenotype including motor neuron disease, cognitive decline resembling frontotemporal dementia, cerebellar ataxia and myopathy" and "ragged-red and cytochrome c oxidase-negative fibres with combined respiratory chain deficiency and abnormal assembly of

complex V" [PMID: 24934289]. Severity is genotype-driven; SMAJ (p.G66V) is a mild lower motor neuron syndrome with normal life expectancy and no mtDNA instability [PMID: 30092269;].

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Quality of life: Per-phenotype QOL instruments specific to FTDALS2 are not available; by extension from ALS/FTD, motor neuron degeneration and cognitive/behavioral decline cause severe progressive loss of daily functioning, with respiratory failure as the terminal event [PMID: 42113599].

4. Genetic / Molecular Information

- **Causal gene:** *CHCHD10* (HGNC:15559; OMIM *615903), chr22q11.23. Small (~14 kDa) mitochondrial intermembrane-space protein with a CHCH domain; intrinsically disordered/low-complexity; heterodimerizes with CHCHD2 [PMID: 35791387;].

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- **Pathogenic variants (representative):**

Variant	cDNA	Class	Phenotype	gnomAD AF	Significance
p.Ser59Leu	c.176C>T	Missense	FTD-ALS (FTDALS2), severe	6.9×10^{-7}	Pathogenic
p.Arg15Leu	—	Missense	Familial ALS/FTD-ALS	0	Pathogenic
p.Gly66Val	c.197G>T	Missense	SMAJ (mild LMN)	2.1×10^{-6}	Pathogenic (Finnish founder)
p.Gly58Arg	—	Missense	Isolated mitochondrial myopathy	—	Pathogenic
p.Ala35Asp	—	Missense	ALS spectrum	6.1×10^{-6}	Likely pathogenic (rare)
p.Pro34Ser	c.100C>T	Missense	(historically ALS)	0.0043	Benign polymorphism (BA1/BS1)

- **Variant type/class:** Predominantly **missense** (dominant). The gene is intolerant to true loss-of-function, but the pathogenic mechanism is dominant missense (Finding 6).
- **Functional consequence:** **Gain-of-function** via toxic protein misfolding, with dominant-negative activity in mammalian systems [PMID: 37021679;

[P 28585542](#)

- **Origin:** Germline (autosomal dominant). No somatic/tumor role.
- **Modifier genes:** The paralog *CHCHD2* modifies mitochondrial function; combined *CHCHD2*/*CHCHD10* loss phenocopies patient mutations, implicating the heterodimer as a functional unit [PMID: 32338760;

[P 30496485](#)

OMA1 and OPA1 are functional effectors/modifiers of the cristae phenotype [PMID: 32338760].

- **Epigenetics / chromosomal abnormalities:** No disease-specific DNA-methylation, histone, or chromosomal (aneuploidy/translocation) findings are established for FTDALS2. mtDNA instability (multiple deletions) is the key acquired genomic lesion [PMID: 24934289].

5. Environmental Information

No specific environmental toxin, radiation, pollutant, occupational exposure, lifestyle factor, or infectious agent has been established as a cause or trigger of FTDALS2. The disorder is monogenic. Oxidative stress worsens the intrinsic mtDNA-repair defect in mutant cells, an inferred sensitizer rather than a primary environmental cause [PMID: 26666268].

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A heterozygous dominant missense mutation in *CHCHD10* (e.g., p.Ser59Leu, p.Arg15Leu) **leads to** production of a misfolding-prone, aggregation-prone mutant protein in the mitochondrial intermembrane space [PMID: 37021679].
2. Mutant CHCHD10 **results in** disassembly of the MICOS complex (MIC60/mitofilin, CHCHD3, CHCHD6) **and thereby** loss of cristae junctions and collapse/dilatation of cristae [PMID: 26666268].
3. MICOS/cristae disruption **leads to** nucleoid disorganization and impaired repair of the mitochondrial genome after oxidative stress, **which causes** accumulation of deleted mtDNA in muscle (mtDNA instability) [PMID: 26666268];

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4. In parallel (branch), mutant CHCHD10 (with loss of the CHCHD2/CHCHD10 functional unit) **activates** the stress peptidase OMA1, **which cleaves** long-form OPA1 (L-OPA1), **further disrupting** cristae architecture [PMID: 32338760].
5. These stresses **induce** the mitochondrial integrated stress response (mtISR) in affected tissues **and result in** respiratory-chain deficiency and abnormal complex V assembly [PMID: 30877432];

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6. Mitochondrial damage **impairs** KIF5B-mediated mitochondrial axonal transport (with syntaphilin dysregulation), **contributing to** synaptic/neuromuscular-junction failure [PMID: 39478664];

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7. Mitochondrial/synaptic damage **drives** cytoplasmic accumulation and aggregation of TDP-43 (co-aggregating with insoluble CHCHD10) — the shared pathological endpoint of ALS/FTLD-TDP [PMID: 28585542;

P 35787294

(The precise CHCHD10→TDP-43 link is partly inferred.)

8. Progressive degeneration proceeds from muscle mitochondrial defect → NMJ degeneration → motor neuron death (demonstrated staging in knock-in mice), together with frontotemporal and cerebellar neuronal loss, **producing** the clinical phenotype of ALS, FTD-like decline, ataxia, and myopathy [PMID: 30874923;

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Branch note: Severity scales with the degree of MICOS disruption/mitochondrial damage rather than with TDP-43 mitochondrial localization [PMID: 30092269]; mild alleles (SMAJ/p.G66V) cause metabolic/creatine-buffering dysregulation without frank mitochondrial myopathy or mtDNA instability [PMID: 40400037;

P 30092269

- **Molecular pathways / cellular processes:** Cristae/MICOS organization; mitochondrial fusion (OPA1) and quality control (OMA1); integrated stress response; apoptosis regulation (mutant fibroblasts inhibit cytochrome c release) [PMID: 26666268]; mitophagy (enhanced in S59L cardiomyopathy) [PMID: 38583639]; mitochondrial axonal transport [PMID: 39478664].

- **Protein dysfunction:** Misfolding/aggregation of an intrinsically disordered protein; dominant-negative disruption of the CHCHD2/CHCHD10 heterodimer [PMID: 35791387;

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- **Metabolic changes:** Dysregulated energy metabolism, altered redox balance, and creatine-based energy buffering [PMID: 40400037].

- **Suggested GO / CL terms:** GO:0042407 (cristae formation); GO:0007007 (inner mitochondrial membrane organization); GO:0140053 (mitochondrial gene expression); GO:0006915 (apoptotic process); GO:0034976 (response to endoplasmic-reticulum/mitochondrial stress) / integrated stress response; GO:0047497 (mitochondrion transport along microtubule). Cell types: CL:0000100 (motor neuron), CL:0000540 (neuron), CL:0000188 (skeletal muscle cell).

7. Anatomical Structures Affected

- **Organ level (primary):** Central and peripheral nervous system — motor cortex, brainstem/spinal motor neurons (UBERON:0001017 CNS; UBERON:0001021 nerve), frontotemporal cerebral cortex (UBERON:0016525 frontal lobe / UBERON:0001871 temporal lobe), cerebellum (UBERON:0002037). Skeletal muscle (UBERON:0001134). Secondary: heart/myocardium in cardiomyopathy-prominent spectrum members (UBERON:0002349) [PMID: 24934289;

P 38583639

- **Body systems:** Nervous (central + peripheral, motor), musculoskeletal, and cardiovascular (spectrum-dependent).
- **Tissue / cell level:** Nervous tissue (upper and lower motor neurons; frontotemporal cortical neurons; cerebellar neurons) and striated skeletal muscle fibres (ragged-red, COX-negative). Cell Ontology: CL:0000100 (motor neuron), CL:0000188 (skeletal muscle cell), CL:0000540 (neuron). The neuromuscular junction is an early failure site [PMID: 30874923].
- **Subcellular level:** Mitochondrion — inner mitochondrial membrane, cristae/cristae junctions, intermembrane space, and nucleoids. GO cellular component: GO:0005743 (inner mitochondrial membrane), GO:0044284 (mitochondrial crista junction), GO:0005758 (mitochondrial intermembrane space), GO:0042645 (mitochondrial nucleoid), GO:0061617 (MICOS complex).
- **Localization / lateralization:** Typically bilateral; may begin focally/asymmetrically as in ALS generally; brain involvement predominantly frontotemporal + cerebellar.

8. Temporal Development

- **Onset:** Adult / late-onset; insidious and chronic. The founding family showed a "*late-onset phenotype*" [PMID: 24934289]. SMAJ presents in mid-adulthood.
- **Progression:** Progressive and neurodegenerative. Rate is genotype-dependent — severe/rapid for FTD-ALS (p.Ser59Leu) versus slow with normal life expectancy for SMAJ (p.G66V) [PMID: 30092269;

P 40400037

Staging (from models): muscle mitochondrial defect → NMJ degeneration → motor neuron death [PMID: 30874923].

- **Disease course:** Chronic, lifelong, progressive; no relapsing-remitting pattern and no spontaneous remission. Duration ranges from a few years (ALS-like) to decades (SMAJ).

Critical intervention windows are inferred to be early (pre-NMJ-degeneration), supported by the muscle-first staging in knock-in mice [PMID: 30874923].

9. Inheritance and Population

- **Inheritance:** Autosomal dominant [PMID: 24934289;

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- **Penetrance / expressivity:** Age-dependent penetrance with variable expressivity; strong allele-specific genotype–phenotype correlation (Finding 7) [PMID: 37021679]. Variant **dosage** modulates severity — the first reported homozygous CHCHD10 (SMAJ) patient had more severe disease, indicating a dose effect [PMID: 40400037].
- **Epidemiology:** FTDALS2 is very rare. CHCHD10 mutations account for ~0.4% of ALS cohorts (2/487 sporadic Chinese ALS) and are a minor contributor to familial ALS/FTD [PMID: 27056076;

P 28318595

Precise prevalence/incidence figures for FTDALS2 are not established (ultra-rare); by comparison, ALS overall affects ~25,000 individuals in the US [PMID: 42113599].

- **Founder effects:** p.Gly66Val (SMAJ) is a Finnish founder allele (gnomAD AF 2.1×10^{-6}) [PMID: 25428574; Finding 6].
- **Population / demographics:** Reported across European (French founding family; Finnish SMAJ) and Asian (Chinese) cohorts. No strong sex bias specific to FTDALS2 is established. Carrier frequency is extremely low given ultra-rare pathogenic allele frequencies (Finding 6).
- **Genetic anticipation / germline mosaicism / consanguinity:** Not features of this dominant missense disorder (not a repeat-expansion disease; not recessive).

10. Diagnostics

- **Clinical / electrophysiology:** EMG and nerve conduction studies to document lower motor neuron involvement (ALS pattern); clinical exam for upper motor neuron signs, cognitive/behavioral (FTD) assessment, and cerebellar signs [PMID: 24934289;

P 42113599

- **Muscle biopsy / histopathology (characteristic):** Ragged-red fibres and COX-negative fibres with combined respiratory-chain deficiency and abnormal complex V assembly — a hallmark distinguishing FTDALS2 from most other ALS/FTD [PMID: 24934289].
- **Laboratory / molecular:** Detection of **multiple mtDNA deletions** in skeletal muscle (mtDNA instability) [PMID: 24934289]; respiratory chain enzymology; metabolomic evidence of altered energy/redox/creatine metabolism (research-stage biomarker) [PMID: 40400037].
- **Imaging:** MRI showing frontotemporal ± cerebellar atrophy (extrapolated from phenotype); no CHCHD10-specific imaging signature established.
- **Genetic testing (definitive):** Single-gene *CHCHD10* sequencing, ALS/FTD gene panels, WES, or WGS. **Variant interpretation is critical** — classify per ACMG/AMP, treating ultra-rare missense (S59L, R15L, G66V) as pathogenic and common variants like p.Pro34Ser (gnomAD AF ~0.4%) as benign (BA1/BS1) (Finding 6) [PMID: 28318595]. Repeat-expansion, karyotype, CMA, FISH, and mtDNA-primary testing are not the diagnostic route (the mtDNA deletions here are secondary to the nuclear CHCHD10 defect).
- **Differential diagnosis:** Other genetic ALS/FTD (C9orf72, SOD1, TARDBP, FUS), primary mitochondrial myopathies with mtDNA-maintenance defects (POLG, TWNK), and SMA. Distinguishing feature: combination of MND/FTD with mitochondrial myopathy + mtDNA instability points to CHCHD10.

11. Outcome / Prognosis

- **Survival / mortality:** Highly genotype-dependent. Severe FTD-ALS (p.Ser59Leu) carries an ALS-like poor prognosis (progressive to respiratory failure), whereas SMAJ (p.G66V) has a normal life expectancy [PMID: 30092269];

P 40400037

FTDALS2-specific survival curves are not established; general ALS survival is ~3–5 years from diagnosis [PMID: 42113599].

- **Morbidity / function:** Progressive motor disability, cognitive/behavioral decline, and myopathy with high burden on daily functioning; respiratory failure is the usual terminal event in ALS-predominant cases.
- **Prognostic factors:** Genotype (specific allele and dosage) is the dominant prognostic factor; degree of MICOS disruption/mitochondrial damage tracks severity [PMID: 30092269];

P 40400037

Metabolic/creatine dysregulation is a candidate prognostic biomarker [PMID: 40400037].

- **Recovery:** None; the disease is progressive and neurodegenerative.

12. Treatment

- **Disease-specific therapy: None approved.** Management follows ALS/FTD symptomatic standards [PMID: 42113599;

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- **Pharmacotherapy (symptomatic, ALS-standard; NCIT terms):** Riluzole (NCIT:C1215; glutamate-release inhibitor; modest ~2–4 month benefit); Edaravone (NCIT:C65358; free-radical scavenger; narrow eligibility). Tofersen is SOD1-specific and **not** applicable to CHCHD10 disease [PMID: 42113599].
- **Supportive / rehabilitative:** Multidisciplinary care (neurology, respiratory support/NIV, nutrition/PEG, physical/occupational/speech therapy) improves survival (~4–7 months) and QOL [PMID: 42113599].
- **Experimental / precision (research-stage):**
- **Nifuroxazide** — repurposed compound rescuing cristae/network abnormalities in S59L patient fibroblasts and reducing caspase-dependent death of S59L iPSC motor neurons via KIF5B-mediated mitochondrial transport enhancement and syntaphilin degradation [PMID: 39478664].
- **Elamipretide/MTP-131** — mitochondria-targeted peptide that enhanced MICOS/OXPHOS in CHCHD2-mutant NPCs (paralog data; candidate for CHCHD10) [PMID: 30496485].
- **Integrated-stress-response** modulation and **metabolic/creatine** support (rationale from mtISR activation and creatine-buffering dysregulation) [PMID: 32338760;

P 40400037

- Broader ALS trials (e.g., trehalose/autophagy in the HEALEY platform) have been negative, underscoring difficulty [PMID: 40409314].
- **Pharmacogenomics / gene / cell / RNA therapy:** No CHCHD10-directed ASO/gene therapy is approved; these are conceptual future directions.

13. Prevention

- **Primary prevention:** Not applicable for a monogenic dominant disorder beyond reproductive options. No vaccine or modifiable risk-factor program exists.
- **Genetic counseling / reproductive prevention:** Autosomal dominant inheritance implies 50% transmission risk; cascade testing of at-risk relatives, prenatal diagnosis, and preimplantation genetic testing are options once a pathogenic variant is confirmed. Accurate variant classification is essential to avoid counseling on benign variants (e.g., p.Pro34Ser) (Finding 6).
- **Secondary/tertiary prevention:** No proven presymptomatic disease-modifying intervention; tertiary prevention focuses on complication management (respiratory, nutrition, falls) within ALS/FTD care standards [PMID: 42113599].

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *CHCHD10* is conserved in mammals; mouse ortholog *Chchd10* (the human-equivalent S59L is modeled as mouse S55L) [PMID: 30877432;

 38583639

The paralog pair CHCHD2/CHCHD10 arose by gene duplication during evolution [PMID: 36158221].

- **Natural disease in other species:** No well-documented naturally occurring CHCHD10 disease in companion animals or wildlife is established (not available in OMIA at time of review). Disease knowledge derives from engineered models.
- **Comparative biology:** The MICOS/cristae machinery and CHCHD10 function are evolutionarily conserved (yeast MICOS mutants recapitulate cristae defects), supporting cross-species mechanistic conservation [PMID: 39478664].
- **Zoonotic potential:** None (non-infectious genetic disease).

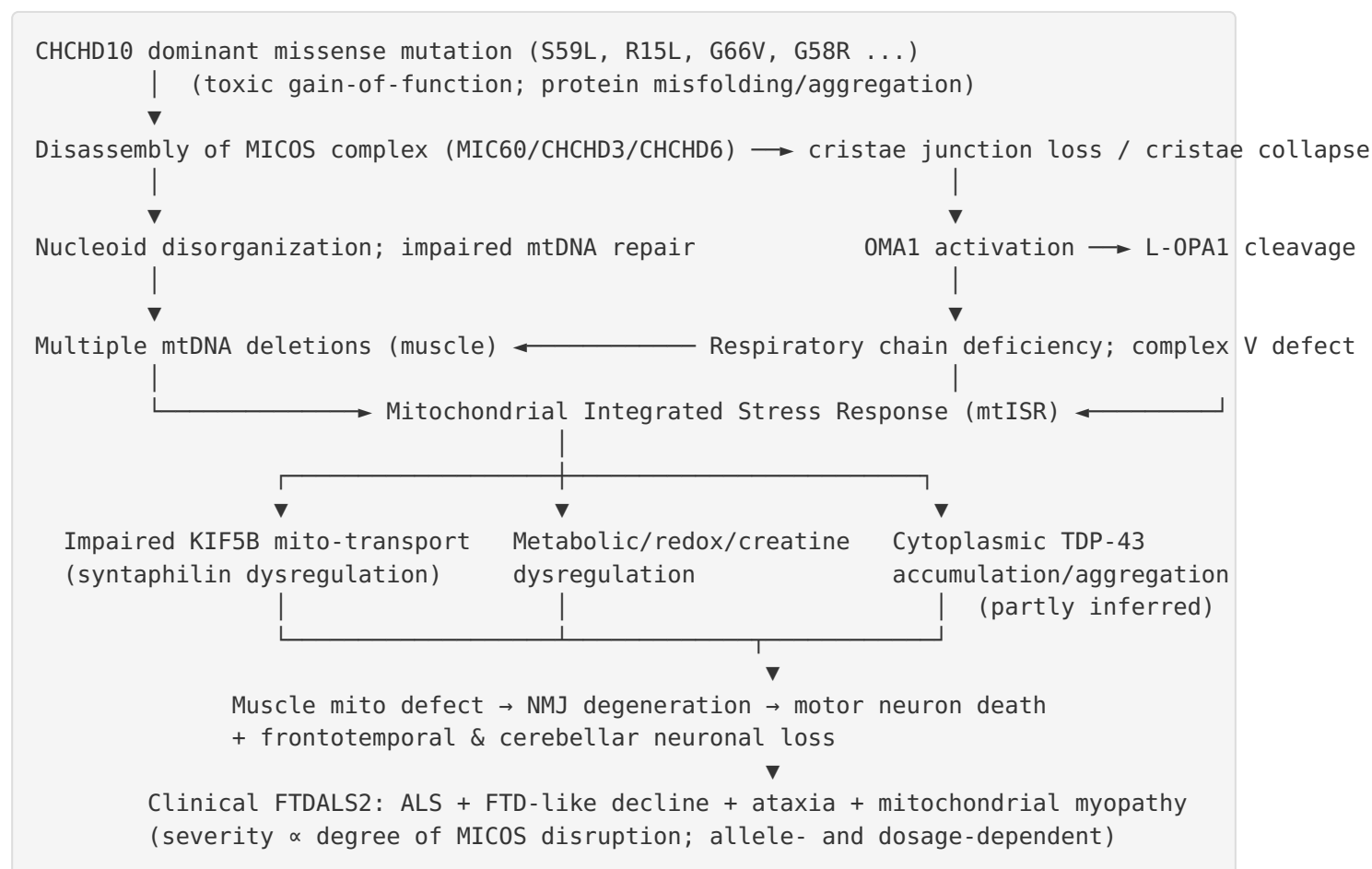
15. Model Organisms

Model	Type	Key features / recapitulation	Reference
S59L / S55L knock-in mouse	Mammalian, knock-in	Tissue-specific toxic gain-of-function; mtISR; fatal mitochondrial cardiomyopathy with enhanced mitophagy; muscle defect precedes NMJ/motor-neuron loss	[PMID: 30877432; P 38583639 ; P 30874923]
C2/C10 double-knockout mouse	Mammalian, KO	OMA1-mediated L-OPA1 cleavage; cristae disruption; cardiomyopathy; mtISR; phenocopies KI mutants	[PMID: 32338760]
G66V knock-in mouse / patient cells	Mammalian + iPSC	Dose-dependent severity; energy/redox/ creatine dysregulation	[PMID: 40400037]
Patient iPSC-derived motor neurons (S59L, G66V)	In vitro (human)	Caspase-dependent death; mitochondrial transport defects; therapy testbed (nifuroxazide)	[PMID: 39478664; P 40400037]
Patient fibroblasts	In vitro (human)	Network fragmentation, cristae abnormalities, impaired mtDNA repair, apoptosis inhibition	[PMID: 26666268; P 39478664]
<i>C. elegans</i>	Invertebrate	Genetic complementation showing LoF-in-complementation + dominant-negative activity; synaptic/TDP-43 phenotypes	[PMID: 28585542]
Yeast MICOS mutants	Cellular	MICOS/cristae biology; repurposing screen platform	[PMID: 39478664]
CHCHD2-mutant hESC/NPC (paralog)	In vitro (human)	MICOS/cristae defects; Elamipretide rescue	[PMID: 30496485]

Model limitations: Mouse KI models show prominent cardiomyopathy that may exceed the human cardiac phenotype; full FTD-like cognitive/behavioral features and the complete ALS motor-neuron degeneration timeline are incompletely captured; TDP-43 pathology recapitulation is partial and its causal link remains under study [PMID: 35787294; 30877432].

P 30877432

Mechanistic Model / Interpretation



The disorder is unified by a single upstream lesion — a misfolding dominant CHCHD10 mutant — that corrupts inner-membrane architecture. The **upstream** events (MICOS disassembly, cristae collapse) are the most severity-determining; **downstream** events (mtISR, transport failure, TDP-43 pathology, neuronal death) produce the clinical picture and connect FTDALS2

to the wider ALS/FTLD-TDP family. The allele determines where on the severity spectrum a patient falls, from mild SMAJ (metabolic dysregulation without frank myopathy) to severe FTD-ALS with mtDNA instability.

Evidence Base

PMID	Title (abbrev.)	How it supports the report
24934289	<i>A mitochondrial origin for FTD/ALS through CHCHD10</i>	Founding family; multisystem phenotype, ragged-red/COX-neg fibres, mtDNA instability
26666268	<i>CHCHD10 mutations promote loss of cristae junctions</i>	MICOS disassembly, cristae/nucleoid loss, impaired mtDNA repair, apoptosis inhibition
32338760	<i>Loss of CHCHD2/CHCHD10 activates OMA1</i>	OMA1→L-OPA1 cleavage mechanism; mtISR; cardiomyopathy
30877432	<i>ALS/FTD mutant CHCHD10 mice: toxic GoF</i>	Knock-in S55L mouse; tissue-specific toxic gain-of-function + stress response
28585542	<i>LoF CHCHD10 mutations, TDP-43, synapses</i>	Dominant-negative activity; cytoplasmic TDP-43 accumulation
30092269	<i>MICOS integrity and severity</i>	Severity $\propto$ MICOS disruption; SMAJ lacks myopathy/mtDNA instability
30874923	<i>Muscle defect precedes NMJ/motor neuron loss</i>	Pathology staging; confirms S59L founding mutation
25428574	<i>LOSMoN/SMAJ caused by CHCHD10</i>	p.G66V mild allele; AD inheritance; spectrum breadth
37021679	<i>CHCHD2/CHCHD10 pathogenesis & precision therapy</i>	Genotype–phenotype map; gain-of-function misfolding mechanism
27056076	<i>CHCHD10 screening in Chinese ALS</i>	Rarity: 0.4% mutation frequency
28318595	<i>CHCHD10 in Mainland China ALS</i>	"Controversial role" — explained by benign vs pathogenic variant confusion
39478664	<i>Nifuroxazide rescues MICOS defects</i>	Experimental therapy; KIF5B/syntaxin transport mechanism
40400037	<i>Dose-dependent CHCHD10 & creatine metabolism</i>	Metabolic dysregulation; dosage-dependent severity; homozygous patient
35787294	<i>CHCHD10, TDP-43 pathology</i>	Insoluble CHCHD10 co-aggregates with phospho-TDP-43 in FTLD-TDP brains
30496485		

PMID	Title (abbrev.)	How it supports the report
	<i>PD-linked CHCHD2 impairs MICOS</i>	Paralog data; Elamipretide rescue; heterodimer biology
36158221	<i>CHCHD2 vs CHCHD10</i>	Heterodimerization; evolutionary duplication
35791387	<i>IDPs in neurodegeneration</i>	CHCHD10 intrinsically disordered; drug-target implications
42113599	<i>ALS: A Review</i>	Standard-of-care therapies; survival; multidisciplinary benefit
42666355	<i>Therapeutic challenges in ALS</i>	Precision-medicine landscape; trial-design context
40409314	<i>Trehalose HEALEY trial</i>	Negative ALS trial; illustrates therapeutic difficulty

Limitations and Knowledge Gaps

- 1. Ultra-rarity limits epidemiology.** No robust prevalence/incidence, sex-ratio, or survival statistics exist specifically for FTDALS2; figures are extrapolated from ALS/FTD generally.
- 2. CHCHD10→TDP-43 causal link is partly inferred.** How mitochondrial dysfunction produces cytoplasmic TDP-43 aggregation is correlative in human tissue and not fully resolved mechanistically [PMID: 35787294].
- 3. Variant interpretation remains a pitfall.** The historical inclusion of benign polymorphisms (p.Pro34Ser) inflated apparent pathogenicity; some rarer variants remain VUS. Systematic functional classification is incomplete.
- 4. Model–human mismatch.** Mouse KI cardiomyopathy may not mirror the human cardiac burden, and cognitive/behavioral FTD features are under-modeled.
- 5. No disease-modifying therapy validated in humans.** All CHCHD10-specific interventions (nifuroxazide, elamipretide, mtISR/metabolic strategies) are preclinical.
- 6. Epigenetic and gene–environment contributions are essentially uncharacterized.**

Proposed Follow-up Experiments / Actions

1. **Curate a CHCHD10 variant registry** integrating gnomAD frequencies, ACMG classification, and phenotype to formally resolve VUS and retire benign misattributions (extends Finding 6).
2. **Functional high-throughput assays** (MICOS integrity, cristae morphology, OMA1/OPA1 cleavage, mtISR readouts) to classify each reported variant along the severity gradient.
3. **Test nifuroxazide and elamipretide head-to-head** in isogenic S59L/G66V iPSC motor neurons and knock-in mice, with mitochondrial-transport and cristae endpoints [PMID: 39478664;

P 30496485

4. **Develop fluid biomarkers** from the metabolic signature (creatine/energy/redox metabolites) for diagnosis and trial stratification [PMID: 40400037].
5. **Mechanistic dissection of the CHCHD10→TDP-43 axis** using proximity-labeling and conditional models to establish causality versus correlation [PMID: 35787294;

P 28585542

6. **Natural-history / registry study** across the CHCHD10 spectrum to define genotype-specific progression and survival, enabling genotype-stratified trial design.
 7. **Explore ASO/gene-silencing approaches** targeting the mutant allele, leveraging the dominant gain-of-function/dosage-dependent mechanism [PMID: 40400037].
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Evidence source types are noted throughout: human clinical (family/cohort studies, autopsy), model organism (knock-in/KO mice, C. elegans, yeast), in vitro (patient fibroblasts, iPSC-derived motor neurons, hESC/NPC), and computational/population-genetic (gnomAD variant frequency analysis).
